

FHS Knowledge and Reality: Modality

Question: Do possible worlds other than the actual one exist?

Core Readings

- David Lewis (1986) *On the Plurality of Worlds* (Blackwell), Ch. 1-3.
- Robert Stalnaker (1976) 'Possible Worlds' in *Noûs* 10(1), pp. 65–75.
- Ted Sider (2003) 'Reductive Theories of Modality' in M. J. Loux and D. W. Zimmerman, eds. *The Oxford Handbook of Metaphysics* (OUP), pp. 180-208.

Further Readings

- Joseph Melia (2003) *Modality* (Acumen)
- Phillip Bricker and Joseph Melia (2008) 'Modality and Possible Worlds' in Ted Sider, John Hawthorne, and Dean W. Zimmerman, eds. (2008) *Contemporary Debates in Metaphysics* (Wiley Blackwell).
- Yagisawa, Takashi (2009) "Worlds and individuals" OUP ch 1-2
- William Lycan (1998) 'Possible Worlds and Possibilia' in Cynthia Macdonald and Stephen Laurence, eds. *Contemporary Readings in the Foundation of Metaphysics* (Blackwell).
- Joseph Melia (1992) 'Against Modalism' in *Philosophical Studies* 68(1), pp. 35–56.
- Gideon Rosen (1990) 'Modal Fictionalism' in *Mind* 99(395), pp. 327–354.
- Kit Fine (1994) 'Essence and Modality' in *Philosophical Perspectives* 8, pp. 1–16.
- See the fac list sections "Transworld identity and Counterpart Theory" and "Essence and Modality" and Stanford articles on "possible worlds" and "counterparts".

Past Year Essay Questions

EITHER

- (a) Are possible worlds properties? If yes, what kind of property? If no, what are possible worlds?

OR

- (b) Why is David Lewis committed to Counterpart Theory? Is such a commitment a weakness of his Modal Realism?

[2025]

EITHER

- (a) Are any two possible worlds qualitatively identical?

OR

- (b) "If you want the theoretical benefits that talk of possibilia brings, the most straightforward way...is to accept such talk as literal truth. It is my view that the price is right..The benefits are worth the ontological cost." (LEWIS). Is this a good reason to accept modal realism?

[2024]

EITHER

(a) Could you have been a poached egg?

OR

(b) Could there possibly have been a plurality of possible worlds? What does your answer tell us about the metaphysics of modality?

[2023]

Can any theory of modality give a plausible account of how we could come to know modal truths?
[2022]

NIL [2021]

What are possible worlds? How do they help, if they do, in accounting for modality? [2020]

PART 0: Basic Problems in Modality

De re and de dicto modality

- De re: What's possible, necessary etc for particular individuals
 - Necessarily, Tibbles is purring or not purring
 - Formalisation: Individual constants inside the scope of modal operators/Quantification into the scope of modal operators
- De dicto modality: What's possible, necessary (etc) but not for particular individuals
 - Possibly, some cat is awake and purring
 - Formalisation: No individual constants inside the scope of modal operators and also no quantification into the scope of modal operators
- Another examples
 - (De dicto) Necessarily, the number of the planets is odd
 - Clearly false
 - (De re) The number of planets is such that it is necessarily odd
 - 9 is presumably necessarily odd (when Pluto was a planet)

Eliminative, Reductive, Non-reductive modal metaphysics

- Eliminative: modal thought and talk is largely somehow defective and it doesn't really latch onto anything in reality (Quine, Melia overview)
- Reductive: modality ultimately boils down to something else; the ultimate story about reality won't explicitly mention modality but won't thereby miss any modal facts
 - Simpler, no unexplained modal principles and simpler epistemology
 - Lewis (Concrete Possible worlds), Armstrong (States of affairs), Fine (Essence), Sider (Conventional choice)
- Non-reductive: ultimate story of reality incomplete if doesn't explicitly mention modality

- Illuminate connections between modal notions, reduce but not eliminate modal primitives, unattractive features of reductive analyses, poor historical track-record for successful reductions
- Possible worlds: Plantinga, Stalnaker

Modality and Epistemology

- Bad Connection: A good metaphysics of modality will make it easier for us to know what's necessary.possible
 - An account of numbers need not make it easier to know truths of mathematics
- Good Connection: good metaphysics of modality should mesh with a good epistemology of modality
 - If we know some modal truth, there should be a good story about how we know it → combining both metaphysics and epistemology

Fundamental Theses

- Necessity is truth at all possible worlds
 - Necessarily, p iff for all w, if w is a possible world, then at w p is true
- Possibility is truth at some possible world
 - Possibly, p iff there exists w such that w is a possible world and at w, p is true
- Fundamental Theses give the following characterisations of possible worlds
 - Maximal: Each possible world supplies a truth value to each proposition: it answers every question
 - Suppose w supplies no truth value to p. Then neither p nor not p is true at w. So p or not p is not true at w. So by the Fundamental Theses, it is not necessary that p or not p. But the laws of logic are necessary
 - Consistent: No possible world supplies more than one truth-value to any proposition
 - Replace above with p and not p is true at w and so possible that p and not p (but contradictions are impossible)
 - Possible Histories: Possible worlds are the various different histories that reality could have had; the various different ways things could have been
 - Possibility concerns the various different ways history could have turned out
 - The Fundamental Theses identify possibility with truth at some possible world
 - So possible worlds are the various different histories that reality could have had

Why Reduction: defining modal notions in terms of the non-modal

- Epistemology: many modal claims are known a priori, and it is a puzzle how this is possible, how we manage to know modal claims without the benefit of sensory experience
 - The epistemology of the modal can be secured if modal notions are defined in terms of notions whose epistemology is secure
- Ontology: reductionism is required by any ontology that claims to give a comprehensive account of reality in terms of primitive entities and notions that do not include modal notions
 - Accepting necessity or possibility as a primitive feature of reality would be like accepting tensed facts as primitive, or accepting dispositions as primitive, or accepting counterfactuals as primitive
- Parsimony: final but probably weakest member
 - Primitivism, the view that modality is unanalysable, is an important and legitimate alternative to reductionism, and is favoured by many because of the difficulty of finding an adequate reduction

PART I: Concrete Realism (Lewisian Theory)

Lewisian Concrete Realism

Fundamental Argument

- P1: The theory of possible worlds enables good accounts of many philosophically interesting phenomena
- P2: No rival theory (or theories) provides a better overall account of those phenomena
- IC: Therefore, we should be more confident in the theory of possible worlds than any rival theory
- P3: The theory of possible worlds entails that possible worlds exist
- C: Therefore, we should be more confident in the existence of possible worlds than any rival anti-realist theory

Sider's overview of Lewisian Concretism

- Reductive account of possible worlds: these possible worlds are taken literally, and are no less real or concrete than actual cows
- Definition: x is a possible world iff (roughly) x is a maximal spatio-temporally interrelated whole, in that (i) any two parts of x are spatio-temporally related to each other and (ii) anything that is spatio-temporally related to any part of x is itself part of x
 - Non-modal definition
 - Lewisian reduction of de dicto necessity: p (where there are no free variables or directly referential terms) is true in a possible world w iff p is true when all

its quantifiers are restricted to parts of w . Then “necessarily, p ” is true iff “ p is true in every possible world w ”

- De re necessity: counterpart theory of de re modal claims
 - To say that Bush is necessarily human is not to say that Bush himself exists in other possible worlds and is human in those worlds; rather it is to say that all of Bush’s counterparts are human
 - No possible individual is (wholly) present in more than one possible world

Concrete Realism

- What kind of entity are worlds: They are concrete universes of just the same kind as our universe
 - Our universe is a maximal connected concrete individual, or spacetime system
 - Maximal: Every part of w is spatiotemporally related only to other parts of w
 - Connected: each part of w is spatiotemporally related to each other part; two things are called worldmates (such as you and I) if and only if they are spatiotemporally related
 - Causally Isolated: Worlds are spatiotemporally and causally isolated. There are no spatial distances or temporal intervals between things in different worlds
 - So: worlds are big concrete entities, all of whose parts are spatiotemporally related to each other and to nothing else
- What is it for p to be true at a world: It is true at w that p iff p is part of w /it is true at w that not p iff p is not part of w
 - Consequence: Whatever there could possibly be, there really is i.e. blue swans are part of some concrete universe
- What is it for an entity to exist or have a property at a world: To exist at w is to have a counterpart at w ; an entity’s counterparts at w are other entities that serve at w as representatives of it.
 - What is true of x at w is what is true of x ’s counterpart(s) at w .
 - A counterpart of x at w is whatever part of w is most similar to x
 - Consequence: different forms of similarity imply different counterpart relations and hence different de re modal truths
 - Why the intuitive solution (the entity is part of the world) fails: nothing is part of more than one world, so nothing exists at more than one world (since w_1 and w_2 are spatiotemporally disconnected)
 - So it follows that if anything have been even slightly different, Tibbles would not exist
 - Kripke’s Humphrey Objection: Although Hubert Humphrey cares very much whether he could have won the election, he “could not care less whether

someone else, no matter how much resembling him, would have been victorious in another possible world”

■ Objection:

- Lois Lane couldn't care less where Clark Kent was born; She does care very much where Superman was born. It doesn't follow that Kent isn't Superman
- Humphrey couldn't care less whether he has an F-counterpart; He does care very much whether he's possibly F. It doesn't follow that having an F-counterpart isn't the same as being possibly F.
- What is it for a world to be actual
 - Actual is a relative notion: w is actual for x iff x is part of w
 - The actual world isn't metaphysically special: it is just the same kind of thing as any other universe, one amongst (infinitely many)
 - Only special for us because we inhabit it
 - Unactualised worlds are relative to where we are → for others in those worlds, we are in an unactualised world

Argument for Premise 1: Concrete Realism and good accounts of philosophically interesting phenomena

- Reduction of modality: modal realism reduces modal talk to (restricted) quantification over concrete entities → possibility just is existential quantification over possible worlds; necessity just is universal quantification over possible worlds
 - “There might be blue swans” is true iff there exists a world W , at W there are blue swans
 - Quantification is restricted to the domain of W : “Possibly, all swans are blue” is true iff there exists a world W , quantifying over and only all its parts, all swans are blue
 - The reason for this is we should not include swans from other worlds (such as the actual world we inhabit) in the quantification – de dicto modal claim
- Semantic Theory for modal logic: modal realism allows us to choose among different relational frames for modal logic
 - Modal Logic
 - Necessarily P is true at w_1 iff P is true at all w_2 such that $w_1 R w_2$
 - Possibly P is true at w_1 iff P is true at some w_2 such that $w_1 R w_2$
 - Lewis thinks that metallurgical results arising from modal logic can help with metaphysical analysis – but this requires us to choose the correct frame R
 - Existence thesis: there exist frames which afford correct interpretations of the modal operators

- Question: what are the properties of the relational frame R? – is it transitive, reflexive? Lewis rightly argues that it requires substantive investigations
 - What kind of worlds are accessible from another, under some kind of necessity i.e. nomological/metaphysical
 - We do not need concrete possible worlds for modal logic – ersatz realism does just as well (sets of sentences); we only need it to choose the axioms
- Example: Every world nomologically accessible from ours – every world that obeys the laws of ours – friction produces heat and so it is nomologically necessary that friction produces heat
 - We might think nomological accessibility to be reflexive, symmetric and transitive → and we get this easily only if we think about it in terms of concrete possible worlds
 - Comes from the substantive investigations about what nomological necessity is
- Systematic theory of relationships between modal notions – simply restrict the scope of quantification over worlds
 - Nomological necessity: Truth at all worlds that obey the laws of our world
 - Historical Necessity: Truth at all worlds that match the history of our world up to a specific time
- Counterfactuals: allows us to
 - Rough idea: “If it were that A, then it would be that C is true” iff C is true at the selected A world
 - Commitment to concrete possible worlds helps us with answering the following questions
 - If one A-world is selected over another A-world, there is a sense in which that A-world is closer to the other – what are the formal properties of this “closeness” ordering
 - Should we write the analysis of conditionals so that it tolerates ties in the similarity relation/tolerates incomparabilities or tolerates a situation in which there are no A-worlds most similar to W, but only more and more similar ones ad infinitum?
 - Is it useful to describe it as a similarity ordering?
 - How should we connect the “would” counterfactual with “might” and probabilistic counterfactuals?
 - Lewis: it is the character of our world that makes the counterfactual true: It is only by bringing the other worlds into the story that we can say in any concise way what character it takes to make what counterfactuals true. The other worlds provide a frame of reference whereby we can characterise our world. By placing our world within this frame, we can say just as much about its

character as is relevant to the truth of the counterfactual: our world is as such to make an (A-and-C) world closer to it than any (A-and-not-C) world is

- Scientific utility: an idealisation of a scientific theory is known to be false at our world, but true at worlds thought to be close to ours such as ideal gases, the ideally rational belief systems, the frictionless plane etc
- Mental content: Content is characterised by a believer's doxastic alternatives – the set of possible individuals who might, for all he believes, be himself
 - Individual X is one of them iff nothing that the believer believes, either explicitly or implicitly, rules out the hypothesis that he himself is X
 - If Pierre believes he is in London, his doxastic alternatives are all individuals who are London-inhabitants in their respective worlds
- Natural Language Semantics: The meanings of sentences are tied to their truth conditions across worlds
 - The speaker, time and world are necessary in determining whether the sentence uttered truly
- Properties, relations, propositions: principal virtue is to avoid accidental co-existensiveness
 - Properties: comes out as the set of instances across all worlds → the property of “being a donkey” comes out as the set of all donkeys i.e. x (at w) satisfies F just when x is part of the extension of F (at w)
 - Avoids accidental co-extensiveness: In our world, “creatures with hearts” and “creatures with kidneys” might be the same set, but they are different properties because there are other worlds where an individual has one but not the other
 - The property does not vary from world to world: what varies is the subset we get by restricting our attention to the world in question i.e. the property of “being a talking donkey” is the same across all worlds, but when we ask “are there x such that x is a talking donkey”, we can no because restricting our attention to this world yields an empty subset
 - Contingency of having a property: Consider Brownie, an otherworldly talking donkey. Brownie himself is, once and for all, a member of the set; hence, once and for all, an instance of the property. But it is contingent whether Brownie talks; Brownie has counterparts who do and counterparts who don't. In just the same way, it is contingent whether Brownie belongs to the set: Brownie has counterparts who do and counterparts who don't. That is how it is contingent whether Brownie has the property.
 - Relations: Dyadic relations are sets of ordered pairs of related things, and we consider all such within-world pairs for all worlds
 - Likewise avoids accidental co-existensiveness

- Propositions: sets of worlds – a proposition holds at a world if that world is a member of that set
- Laws of Nature
 - Supervenience: we have supervenience when there *could* be no difference of one sort without differences of another sort
 - Example: A dot-matrix picture has global properties - it is symmetrical, it is cluttered, and whatnot - and yet all there is to the picture is dots and non-dots at each point of the matrix. The global properties are nothing but patterns in the dots. They supervene: no two pictures could differ in their global properties without differing, somewhere, in whether there is or isn't a dot.
 - Example 2: The world has its laws of nature, its chances and causal relationships; and yet - perhaps! - all there is to the world is its point-by-point distribution of local qualitative character. We have a spatiotemporal arrangement of points. At each point various local intrinsic properties may be present, instantiated perhaps by the point itself or perhaps by point-sized bits of matter or of fields that are located there. There may be properties of mass, charge, quark colour and flavour, field strength, and the like; and maybe others besides, if physics as we know it is inadequate to its descriptive task. Is that all? Are the laws, chances, and causal relationships nothing but patterns which supervene on this point-by-point distribution of properties? Could two worlds differ in their laws without differing, somehow, somewhere, in local qualitative character?
 - Lewis: simple solution to supervenience doesn't avail itself to people who think all modality must come in diamonds and boxes (i.e. just some formal notion)
 - Some people: read "could" in supervenience as diamond operator i.e. It is not the case that there is some world W such that, at W, two things have a difference of the one sort but not the other
 - Problem with broad psychological supervenience: could two people differ mentally without there being a physical difference somewhere, whether in the people themselves or somewhere in their surroundings?
 - The thesis we want says that there could be no mental difference between two people without there being some physical difference, whether intrinsic or extrinsic, between them
 - BUT if we take 'could' as a diamond: **there is no world** wherein two people differ mentally without there being some physical difference, intrinsic or extrinsic, between them → there could

be no mental differences without some physical difference of the sort that could arise between people in the **same world**.
(we only look at differences in the same world)

- We ignore extrinsic physical differences that only ever arise between people in **different worlds** i.e. differences between two people existing in different geometries

Epistemology of Lewisian Concrete Realism

Problem (Benacerraf's Dilemma): How can we have knowledge of things (like sets or worlds) that we cannot causally interact with

- But we have abundant modal knowledge: we know that there might be a talking donkey etc
- Just like mathematics, our knowledge about modal claims is much more secure than our knowledge about epistemology – to reform modality to serve epistemology seems suspect
 - We can know mathematical claims even though they are causally inert
- But the analogy with mathematics fails
 - Skyrms: If possible worlds are supposed to be the same sorts of things as our actual world; if they are supposed to exist in as concrete and robust a sense as our own; if they are supposed to be as real as Afganistan, or the center of the sun or Cygnus A, then they require the same sort of evidence for their existence as other constituents of reality
 - If possible worlds were exist in some abstract sense, then they would not require the same kind of evidence

Lewis's solution

- Denial of Causal Acquaintance: Causal accounts of knowledge cannot be general theories of knowledge, because our knowledge of mathematical claims refutes them
 - Whether or not causal acquaintance is required is not demarcated by concreteness of the entity in question, but of contingency
 - If I know by seeing, for instance, my visual experience depends on the scene before my eyes; if the scene had been different, within limits, my experience and my subsequent belief would have been correspondingly different.
 - But nothing can depend counterfactually on non-contingent matters i.e. on what possibilities there are
 - The antecedent "if there were no possibility for dragons and unicorns to exist in a single world" is impossible, and so the counterfactual is vacuously true

- A priori knowledge: if we are prepared to expand our existential beliefs for the sake of theoretical unity, and if thereby we come to believe the truth, then we attain (modal) knowledge.
 - This allows us to come to know that there exist countless objects causally isolated from us and unavailable to our introspection
 - Our knowledge that there are donkeys at some worlds is necessary (our knowledge that there are donkeys in our world is contingent)
 - **Principle of Recombination** (to achieve a plenitude of possible worlds): anything can coexist with anything else, at least provided they occupy distinct spatiotemporal positions, size and shape permitting; likewise, anything can fail to coexist with anything else
 - If there could be a dragon, and there could be a unicorn, but there couldn't be a dragon and a unicorn side-by-side, that would be an unacceptable gap in logical space
 - What about differences with our laws of nature? There could be something travelling at the speed of light?
 - The only limit on the extent to which a world can be filled with duplicates of possible individuals is that the parts of a world must be able to fit together within some possible size and shape of spacetime
 - Alternative Principle of Recombination: for any collection of entities, there is a world containing duplicates of those entities in any spatio-temporal configuration whatsoever
 - Intuitively, every way of distributing possible objects over spacetime corresponds to some possible world

Ersatz Modal Realism

Linguistic Ersatzism: Maximally consistent sets of sentences

- Represent by saying: If the worldmaking language is plain English, for instance, then it is the case according to a novel that a donkey talks iff that novel contains the sentence 'A donkey talks'.
- Cheap ontology: ersatz worlds turn out to be set-theoretic constructions out of words. A word in turn might be taken as the set of its particular inscriptions; or as the set of spatiotemporal regions wherein it is pronounced. Thus the ersatz worlds are set-theoretic constructions out of parts of the concrete world.
- Problem of Cardinality: if we take "language" literally, so that sentences are finite strings over a finite alphabet, there are not enough sets of sentences to go around → there are at most a continuum amount of sets of sentences (easy to see) and hence at most a continuum amount of ersatz worlds
 - But it is at least possible to have continuum many spacetime points; each point may be occupied by matter or may be vacant; since anything may

coexist with anything, any distribution of occupancy and vacancy is possible; and there are more than continuum many such distributions.

- But can not take language literally → language in a generalised sense is the only thing that is needed i.e. a system of structures that can be parsed and interpreted
- Primitive Modality: cannot reduce modal talk
 - Via consistency: an ersatz world must be consistent i.e. must be a consistent set whose consistency would be destroyed by the addition of any other sentence of the worldmaking language.
 - A prima facie modal distinction between acceptable and unacceptable ersatz worlds: a set of sentences is consistent iff those sentences, as interpreted, *could* all be true together
 - Implicit Representation of a statement:
 - It may be that so-and-so, according to a certain ersatz world, not because there is a sentence included in that world which just means that so-and. so, no more and no less; but because there are sentences which jointly imply that so-and-so. There might be a single sentence which implies that so-and-so but doesn't just mean that so-and-so because it implies more besides; or there might be a finite or infinite set of sentences which jointly imply that so-and-so.
 - The implication is prima facie modal: a set of sentences implies so-and-so iff those sentences, as interpreted, could not all be true together unless it were also true that so-and-so; in other worlds, if it is necessary that if those sentences are all true together, then so-and-so
- Alien Properties: if a worldmaking language is specifiable by a this-worldly theorist, then it cannot be expected to have a rich enough vocabulary to describe and to distinguish all different possibilities there are
 - If we build an ersatz worlds out of actual things, linguistically or otherwise, how can we represent possibilities that involve extra individuals
 - Problem about naming → Lewis rejects that this is a problem (i.e. how can we distinguish possible worlds if we have no language to name extra individuals) due to his counterpart theory
 - Problem for properties: if we only have words for the natural properties that are instantiated within our actual world, then we are not in a position to describe completely any possibility in which there are extra natural properties, alien to actuality.

Pictorial/Magical Ersatzism

- A pictorial ersatz world represents the concrete world by being a "picture in a generalised sense"

- Pictorial ersatzism treats worlds as being like **pictures, scale models, or three-dimensional diagrams**
 - Possible worlds are treated as **abstract representations** that represent by isomorphism
- Just as a picture of a cat on a mat has a part that corresponds to the paws and a part that corresponds to the whiskers, an ersatz world consists of abstract parts with diverse properties arranged in a specific way
 - One ersatz world pictures the concrete world correctly; this is the **actualised** ersatz world
 - Any other ersatz world is unactualised because it misrepresents the world—it represents the world as being isomorphic to *it*, but the concrete world is actually isomorphic to something else
- Key advantage: handles alien properties and need not name them to represent them
 - A pictorial ersatzer can simply claim that their abstract pictures contain parts that instantiate these alien natural properties
- Primitive Modality: the set of possible worlds
 - The ersatzer must explain why only *possible* pictures count as worlds. For example, why is there no pictorial world where a particle is both positive and negative in charge?
 - If the coexistence of such charges is impossible, then no such picture exists; but to say this, the ersatzer must rely on a primitive modal fact about what is possible, rather than reducing modality to existence as Lewis does
- Primitive Modality II: Implicit Representation
 - A picture represents "explicitly" only the properties shared across the divide between abstract and concrete
 - However, complex properties (like "being a talking donkey") are represented implicitly
 - A picture represents a talking donkey if it represents a specific arrangement of atoms and parts that *implies* a talking donkey
 - Lewis argues this implication is itself a modal notion: "necessarily, if the parts are thus-and-so, then there is a talking donkey"
- Vim Objection: Lewis's most serious objection is that pictorial worlds are not truly "abstract" in a way that makes them easier to believe in than his concrete worlds
 - He argues that if these pictures are isomorphic to concrete worlds in every detail, then an abstract donkey in an ersatz world is just like a concrete donkey, except it lacks a "mystery ingredient" Lewis calls **vim** (his term for concreteness or actualisation)
 - If the only difference between a "concrete" world and an "abstract" pictorial world is this invisible quality of vim, then the ersatzer's ontology is no more "common-sensical" than modal realism; it just involves an infinity of vimless donkeys instead of an infinity of donkeys with vim

Brief Overview of Counterpart Theory

Core Arguments and Overview Lewis rejects trans-world identity (the idea that one individual exists in multiple worlds) to avoid the problem of accidental properties.

- The Problem of Overlap: If Hubert Humphrey existed in two worlds, and in one he has five fingers while in the other he has six, then Humphrey (who is self-identical) would have both five and six fingers, a contradiction.
- Counterpart Relation: Lewis proposes that an individual exists in only one world but has counterparts in others. The counterpart relation is one of similarity (in origins, character, or role) rather than identity.
- Mechanism: A world represents Humphrey *de re* (e.g., "Humphrey might have won") not by containing him, but by containing a counterpart of him who won.

Relationship to Concrete Realism Counterpart theory is the "favoured" way for a modal realist to handle *de re* modality while maintaining that worlds are isolated and do not overlap. While modal realism itself is primarily about quantification over worlds, Lewis considers counterpart theory essential for providing a coherent, non-contradictory account of how individuals have potentialities and essences without living "double lives" in different worlds

Criticisms of Lewisian Theory

Objection Class 1: Costly Ontology

- The Incredulous Stare
 - Lewis follows Quine's conception of ontological commitment: according to Lewis, his theory of possible worlds provides the best systematic account of modal and other phenomena; its ontology is therefore reasonable
 - Parallels Quine's argument for the existence of sets: our best overall empirical theory, mathematical physics, quantifies over real numbers; therefore, we have reason to posit real numbers, or the sets to which they may be reduced
 - See indispensability arguments
- Ontological Profligacy
 - Basic argument
 - P1: Theories with large ontology are bad, other things being equal
 - P2: Concrete realism's ontology is really large
 - C: Concrete realism is bad, other things being equal
 - Lewis's reply
 - Quantitative parsimony (how many things) unimportant; qualitative parsimony (how many primitives) more important
 - New argument
 - P1: If concrete realism is true, then whatever kinds there could be, there are and whatever entities could be, there are

- P2: Whatever kinds there could be, there really are is maximally qualitatively unparsimonious; Whatever entities there could be, there really are is maximally quantitatively unparsimonious
- C: Concrete realism is maximally unparsimonious qualitatively and quantitatively
- Reply
 - Lewis's ontology is not larger than required (only ontology larger than required is bad)
 - Williamson: parsimony is the least important; what matters overall is complexity, strength and explanatory power
 - Lewis: clearly not all things equal because concrete realism is better than all rivals

Objection Class 2: Epistemic Objections

- Peacocke's Integration Challenge
 - Metaphysics and epistemology need to mesh: epistemology should provide a good story about how we come to know commonplace modal claims such as "We could all have stayed in bed today"
 - Generalisation of Benacerraf's problem for mathematical knowledge: how might we explicate mathematical truth in a way that would "fit into an overall account of knowledge in a way that makes it intelligible how we have the mathematical knowledge we have"
 - Applies to abstract realists too: unclear why intuitions or conceivability should give us any evidence about mind-independent metaphysical modal facts
- Objection from Causal Acquaintance
 - See Above
- Skepticism about Recombination I: Fritz's A Purely Recombinatorial Puzzle
 - Naive combinatorial reasoning leads to an inconsistency itself
 - Needs to be restricted by some essentialist claims such as "impossible for there to be Socrates without there being Socrates's parents"
 - But usually not in conflict such as having a dragon and a unicorn side by side
 - Puzzle: Expansion Principle and the Plenitude Principle
 - Expansion Principle: The following is necessarily the case: whatever angels there are, it is possible for all of them to be angels and for there to be an angel which is actually not an angel
 - Consider w. Take all things in w which are angels in w. By Recombinatorial reasoning, there is then a possible world v in which all of them are angels, and in which there is an extra angel i.e. in which there is something which is an angel in v but is not an angel in w

- Motivated: recombinatorial idea of being able to duplicate a given individual to generate a further possibility
 - Plenitude Principle: Possibly, all possible angels are angels
 - There is a possible world in which all possible angels are angels there
 - Generally: there is a possible world that contains **all possible individuals** of a certain kind
 - - Motivation: by recombinatorial reasoning, there should be a world in which all possible angels are angels i.e. take all individuals x such that for some world, x is in that world and x is an angel
 - But the Expansion Principle and Plenitude Principle come into conflict (when formalised in an enriched first-order modal logic) even though we judge them individually to be correct
- If Lewis is right and the Principle of Recombination is just like a priori reasoning in mathematics, then it shouldn't lead to some sort of contradiction!
- Reliability Challenge (Nozick)
 - If we are to be justified in believing something in a given domain, we must have a reliable faculty for forming beliefs of that sort, the existence of which is best explained in terms of natural selection
 - Nozick: where modal beliefs are concerned, "we do not appear to have such a faculty, and it is implausible that evolutionary processes would instill that within us"
 - Since we have no good explanation of why we should have developed a reliable faculty for detecting metaphysical necessity, we should be skeptical about claims that we have such knowledge, and even about the claim that there are such necessities
 - Vetter's Reply: modal knowledge is knowledge of modal properties of this world such as potentialities and dispositions of actual objects – undercutting the feeling of inaccessibility
 - Williamson's Reply: ordinary cognitive capacity to handle counterfactual conditionals carries with it the capacity to evaluate metaphysical modality. This cognitive capacity carries with it an evolutionary story
 - Metaphysical counterfactual thinking is logically equivalent to a special case of counterfactual thinking, since a proposition is necessary if its negation counterfactually implies a contradiction
 - But: how do we come to know counterfactual conditions? Williamson has a story about supposing an antecedent condition and develops

- the supposition by adding further judgements within the supposition by reasoning, offline predictive mechanisms and offline judgements
 - But: Williamson's solution clearly not generalizable to all counterfactual conditionals
 - But: Nozick is clearly arguing for skepticism about our knowledge claims about distinctively metaphysical necessities and possibilities; these metaphysical claims are not analogous to claims about whether the rock would have rolled into the lake
 - Instead: "If Mary's brain were successfully transplanted into another body, would Mary survive"
- Skepticism about Recombination III: Thomassen
 - Disanalogy with mathematical truth: mathematical truth consists of abstracta while Lewisian worlds, by hypothesis, concern concreta
 - About Belief rather than knowledge: Lewis offers a story only about how we in fact come to hold the modal beliefs we do – by way of general reasoning from principles of recombination, guided by imaginative reasoning.
 - But: impossible to see why these principles of recombination, given modal realism, should be any good whatsoever at revealing to us what is going on in these independent, spatiotemporally and causally isolated worlds
 - Vetter: it seems like black magic that the epistemology of Recombination should just happen to get the metaphysics of possible worlds right, given that there is no connection by which the latter might have informed the former
- Skepticism about Recombination IV: Metalogical principles
 - Principles of Necessary Existents, Barcan Formula etc
 - Lewis: Concrete Possible Worlds gives us the background understanding to choose among relational frames to represent different necessities (such as S5)
 - But: it seems unclear how recombinatorial reasoning could tell us such general, meta-facts about possible worlds i.e. choosing between a simple, constant domain quantified modal logic, or a variable domain variant

Objection Class 3: Metaphysical Considerations

- Objection from Actuality
 - Actualism: the claim that it is a conceptual truth that everything is actual
 - Lewisian Possibilism: existence of non-actual things
 - Actualist: possibilism is conceptually incoherent since 'actual' is simply a blanket term for absolutely everything
 - If Lewisian worlds really did exist, they would simply be part of actuality and would have nothing to do with possibility and necessity

- Lewis's reply: consider three claims
 - (1) Everything is actual; (2) actuality consists of everything that is spatio-temporally related to us and nothing more. (3) possibilities are not parts of actuality, they are alternatives to it
 - Lewis denies (1) and commits to (2) → doesn't see any claim as substantively more compelling or that analyticity is concentrated more in some of them than in others
 - Sider: linguistic convention supplies defeasible constraints on meanings i.e. constraints that may be defeated by what candidate meanings exist
 - E.g. convention supplies the defeasible constraint that things that are in contact have absolutely no space between them; in fact, there is no relation of this sort between material bodies. As a consequence, the meaning of the term "contact" is a relation that does not satisfy this convention constraint → so in a sense, "things that are in contact have absolutely no space between them" is analytic but false
 - False because it is outweighed by other considerations (i.e. by physics) and analytic only in virtue of linguistic but defeasible constraints
- Island Universes
 - Lewis's definition of a possible world as a spatio-temporally interrelated whole rules out the possibility of an absolutely empty possible world (but allows the possibility of a world containing nothing but empty spacetime)
 - Rules out the existence of a single world containing two disconnected space time → intuitively possible
 - Argument
 - P1: Reality could have contained some spatiotemporally unconnected island universes
 - P2: If concrete realism is true, it is impossible for there to be some spatiotemporally unconnected island universes
 - restricting attention to parts of w, the w is a maximally spatiotemporally interrelated whole/spacetime
 - C: Concrete realism is false
 - Sider's Response (Redefine possible world): Suppose Lewis counted every individual as a possible world – even individuals containing parts from different maximal spatiotemporally interrelated sums.
 - Possible worlds would overlap extensively i.e. I am a possible world as would the aggregate of me plus the Eiffel tower etc etc

- Now consider two maximal spatio-temporally interrelated sums – two Lewis worlds → on this new definition, the sum of these two possible individuals (which are disconnected) itself counts as a possible world
 - Sider's caveat: on this view I inhabit a possible world that contains a talking donkey since we can take a fusion of me, any possible talking donkey, and zero or more other things → but we do not want an everyday utterance "there exists a talking donkey" to be true → add to the view the claim that in non-modal sentences, quantifiers are typically restricted in some way
 - Bricker's Solution (Redefine Modality): modal operators should be taken to be quantifying over worlds and pluralities of worlds. To be possible is to be true at some world, or some plurality of worlds
 - Isn't this just expanding the definition of world to include disconnected spacetimes? I.e. a plurality of spacetimes
 - See more below
- Advanced Modalising
 - Lewis's definition of possible worlds is such that there is only one world present at each w. So it is true in each world that there's only one spacetime system. Hence, it is necessary that there's only one spacetime system
 - Argument
 - P1: If concrete realism is true, it's impossible that there's a plurality of spacetime systems
 - P2: What's impossible is not true
 - P3: If concrete realism is true, there isn't a plurality of spacetime systems, and so concrete realism is false
 - C: Concrete Realism is false
 - Cian Dorr's How to be a Modal Realist: See Below
 - Diver's Theory of Advanced Modalising
- Shalkowski's Objection
 - Hidden circularity: Lewis must require that there exists a Lewis-world for every possible way things could have been
 - Imagine that Lewis worlds containing 9-feet-tall humans do not exist → then certain propositions will incorrectly turn out impossible → to rule out gaps in logical space
 - Sider: Suppose that reality is just the way Lewis thinks it is i.e. that Lewis's multiverse exists and there is a possible world for all and only the possible ways things might have been
 - Is there a candidate property we can identify with the property of being a possible proposition? Answer seems to be yes – it is the property of being true at some Lewis world
 - Actually non-reductive

- Then there is a question of whether it is reasonable to believe that Lewis's ontology is correct → Lewis has his Quinean answer → believe because of its theoretical utility

Bricker's Island Universes and Analysis of Modality

Key Claim: we amend our analysis of modal operators to evaluate against possible worlds and a plurality of possible worlds

- Problem
 - According to Lewis, individuals are part of the same possible world **if and only if they are spatiotemporally related**.
 - If a possible world must be spatiotemporally unified, then by definition, no single possible world can contain island universes (spatiotemporally isolated parts)
 - Too narrow a consequence: Under the standard analysis where "possible" means "true at some possible world," island universes are rendered **metaphysically impossible**
- Bricker's argument for plenitude: endorses several principles
 - **The Principle of Solitude (PS):** Bricker asserts that anything that can exist can exist "all by itself". If we take a world with two separate regions and remove the intervening space-time, those regions should still be able to exist as isolated duplicates
 - **The Generalized Principle of Solitude (GPS):** Bricker argues there should be no arbitrary restrictions on what can be actualized. If God (or a primitive actualization process) can choose to actualize one world, there is no logical reason why God could not choose to actualize **two or more worlds simultaneously**, or even all of logical space
 - **Accommodating Intuition:** Bricker wants a theory that can handle other "troublesome" possibilities that Lewis rejects, such as the **possibility of nothing** (no physical reality existing at all) rather than merely empty spacetime
- Solution: Plural Quantification
 - Instead of analyzing possibility as truth at a single possible world, Bricker proposes that **modal operators be analyzed as plural quantifiers over possible worlds**
 - A proposition is metaphysically possible if it is true at some world **or some plurality of worlds**
 - Under this analysis, the proposition "island universes exist" is possible because it is true when evaluated against a **plurality of two or more spatiotemporally isolated worlds**. This allows the realist to keep the narrow definition of a "world" (as a unified individual) while expanding "possibility" to include combinations of those worlds

- Caveat: ditching the indexical analysis of actuality
 - Semantic failure: If someone asserts, "Island universes exist," an indexicalist must ask: **relative to which class of worlds is this utterance evaluated?**
 - If truth is evaluated only relative to the single world the speaker inhabits, then "island universes exist" is **analytically false** because a single world, by definition, has no absolutely isolated parts
 - If one tries to evaluate the utterance relative to *any* plurality containing the speaker's world, there is "nothing to choose between them," leading to the conclusion that the statement is **neither true nor false**
 - Bricker endorses **absolute actualization**, which holds that there is an absolute ontological distinction between what is actual and what is merely possible
 - Actuality is metaphysically privileged in some way
 - But: he does not say that this world in which we inhabit is the only world but instead the only world we are causally acquainted with
 - So: An assertion like "island universes exist" is evaluated relative to **the class of all worlds that have been absolutely actualized**

Cian Dorr's How to be a Modal Realist

Key Claim: Considers problems in advanced modalising and argues that modal operators are extraneous in unrestricted quantification

- Problem
 - Lewisian View
 - By definition, it is possible that some swans are blue iff there is a Lewis world such that some swans in it are blue
 - Can replace "by definition" with "necessarily"
 - Puzzle 1
 - Necessarily: it is possible that some swans are blue iff there is a Lewis world such that some swans in it are blue
 - But: it is possible that no swans are blue and it is also possible that there be contingently no blue swans (i.e. possible no swans are blue even though it is possible that some swans are blue)
 - If it is possible that there be contingently no blue swans, it would have to be possible that no swans are blue even though some swans in some Lewis world are blue.
 - Contradiction: any blue swan that is part of a Lewis world is a blue swan → denies that there are contingently blue swans
 - Puzzle 3
 - Necessarily, it is possible that Q Fs are Gs iff there is a Lewis world such that Q Fs in it are Gs

- Q is some quantifier such as “some” or “many” or “at least two”
 - Necessarily, it is possible that at least two things are Lewis worlds iff there is a Lewis world such that at least two things in it are Lewis worlds
 - But: by definition of a Lewis world, there is only one world in any Lewis world, and hence we have to say it is not possible that at least two things are Lewis worlds
 - But: there are infinitely many Lewis worlds!
- Restricted quantification: treat Q as restricted to the domain of our world
 - Puzzle 1: ‘possibly some_r swans are blue’ is not inconsistent with ‘no_r swans are blue’ and so consistent with “Possibly, there are contingently no_r blue swans”
 - Puzzle 3: All the statement tells us is that it is not possible that there be at least two_r Lewis worlds. It follows by T (truth → possibility) that there is at most one_r Lewis world. But this is consistent with Lewis’s claim that there are infinitely many_u Lewis worlds
- Unrestricted quantification: how to analyse “Possibly Q Fs are Gs” when Q is taken as unrestricted
 - Principle of compositionality: it is possible that some swans are blue also admits of both kinds of readings (since “some” itself admits both restricted and unrestricted readings)
 - Dorr: best option to hold that “Possibly Q_u Fs are Gs” is equivalent to every case of “Q_u Fs are Gs”
 - So “Possibly that there are a plurality of spacetime systems” → “There are a plurality of spacetime systems” i.e. ignoring the at-a-world evaluation
 - “Q_u Fs are Gs” → “Possibly, Q_u Fs are Gs” by T; “Possibly, Q_u Fs are Gs” → “Q_u Fs are Gs” by common sense
 - Why should we not restrict Q_u to some appropriate part of reality i.e. “Possibly, Q_u Fs are Gs iff some_u appropriate part of reality is such that Q_u Fs in it are Gs”
 - One reason: the whole of reality better count as an appropriate part to preserve validity of T (i.e. truth implies possibility)
 - Another: If anything short of reality as a whole counts as “appropriate”, we will no longer have an S5 modal logic, even for claims involving just unrestricted quantification and qualitative predicates
 - If there is an appropriate part that lacks blue swans, then it is possible that no_u swans be blue, and hence possible that it is not possible that some_u swans are blue

- But the modal realist is already committed to rejecting common-sense modal claims: the claim that it is possible that it is contingently the case that no_u swans are blue

Divers' A Genuine Realist Theory of Advanced Modalizing

Key Claim: Advanced possibly claims are redundant

Advanced Modal Claims: the problem

- Advanced Modal Claims: Modal claims about entities other than spatiotemporally unified individuals (i.e. spatiotemporally disunified individuals, sets, numbers, properties etc)
- Principles that any theory must accommodate: pre-theoretical data
 - Modal Ubiquity (MU): For every true statement p, it is true that it is necessary that p or it is true that it is contingent that p
 - Possibility Introduction (PI): it is possible that p is a semantic consequence of p
- Problem 1:
 - (1) There are many worlds
 - (P1) It is possible that there are many worlds
 - But (P1) will be found false because by definition of the Lewis world, there is exactly one Lewis world at each w
 - (2) There are natural properties
 - (P2) It is possible that there are natural properties
 - Natural properties are invariably transworld according to Lewis, and so are never wholly in one world as (P2) supposes
- Problem 2: Skyrms's superworld
 - Superworld (S): A genuine modal realist must recognise a "superworld" into which all possible worlds are collected
 - This is because a genuine realist does recognise the unrestricted sum of all individuals, of which every (possible) world is a proper part. Call this world b
 - Lewis's Truism (LT): All things could have been otherwise
 - But S and LT are inconsistent because the superworld b could not have been otherwise since it includes all the possibilities
- Problem 3: Yagisawa's accessibility relation
 - Contingent Inaccessibility (CI): w could have been inaccessible from w'
 - Yagisawa's observation: content of modal statements about big possibilia (e.g. worlds) should be given in the same way as that of modal statements of smaller possibilia (e.g. Humphrey). Size shouldn't matter and ordinary modalizing about a world should be dealt with counterpart-theoretically

- But clearly CI cannot be accommodated: there is no world w'' such that w is inaccessible from w' because the “accessibility relation” is a transworld phenomena/relation

Diver’s Redundancy Solution

- Weakening MU and trivalence: if we detect truth value gaps at the advanced modal claims, we can say
 - MU*: For every true statement p , it is not true that (it is not necessary that p and it is not contingent that p)
 - We have $\{0,1,\#\}$
 - But sacrifices bivalence
- World-inclusion relations: further species of world inclusion relations beyond being wholly in a world i.e. *being partly in a world or existing from the standpoint of a world*
 - An entity exists from the standpoint of a world iff “it belongs to the least restricted domain that is normally – modal metaphysics being deemed abnormal – appropriate in evaluating the truth at that world of quantifications” (Lewis)
- What world-inclusion relation solutions fail
 - When we move from “There is some x such that $A(x)$ ” to “It is possible that there is some x such that $A(x)$ ”, the modal expression does not have the semantic effect of altering the range of formerly restricted quantifiers that now occur within its scope (i.e. still restricted rather than unrestricted)
 - For transworld modal claims or advanced modal claims, reference to transworld entities or the articulation of transworld states of affairs involve quantification that is not world-restricted
- Solution: Redundancy
 - In cases where the quantifiers were not formerly world-restricted, the possibility operator has no semantic effect on the content of the sentence within its scope → semantically redundant
 - Whenever possibility operator expresses a non-trivial semantic function on quantificational sentences, it is always altering the scope of the formerly world-restricted quantifiers
- How the solution solves
 - Problem 1: “Possibly, there are many worlds” is just “There are many worlds” where the quantifier “many” is taken to be unrestricted
 - PI is preserved since $p \models p$ is always valid
 - Problem 2: Skyrms’s superworld
 - Superworld b could have been otherwise is just in respect to some F : F_b and it is possible that not F_b
 - By redundancy: F_b and not F_b → so we reject the existence of Skyrms’s superworld systematically

- Problem 3: Yagisawa's inaccessibility relation
 - It is possible that w is inaccessible from w' just as w is inaccessible from w' , contradicting the hypothesis

Yagisawa's Extended Modal Realism

Key Claim: Modality is another dimension of reality, like space and time. "Possible worlds" are identified with another point along the modal dimension of reality, and include impossible worlds

Motivation

- Parallel between temporal and modal logics: Yagisawa's primary motivation is the **extensive parallel between temporal and modal logic**. He argues that just as we use temporal tense to distinguish between what is happening *now* and what *was* happening, modal realists should employ **modal tense** to distinguish between what is happening *actually* and what *is merely possible*.
- Rejection of Constituent Modal Realism: In constituent realism, possible worlds are seen as **containers** or mereological sums of individuals. Yagisawa thinks this goes wrong in two major ways:
 - **The Problem of Representation**: Yagisawa argues that Lewis's **counterpart theory** is "less realist in an important sense". To say "Humphrey could have won" by pointing to a different individual (a counterpart) who wins in another world is, for Yagisawa, to use a **representation** rather than the *res* (Humphrey himself)
 - **Modal Chauvinism**: Actualists (and to an extent Lewis) treat the actual world as metaphysically privileged. Yagisawa seeks a "metaphysically egalitarian perspective" where the non-actual is as real as the actual, modeled after **four-dimensionalism** in time, where past and future individuals are as real as present ones

Theory: Possible worlds as points (or indices) along the modal dimension

- In Yagisawa's construction, the universe does not merely exist in space and time; it **spreads out in a modal dimension**
- What is a World? A world is not a concrete object or a container; it is a modal index (a point or location in modal space), just as a moment is an index in the temporal dimension. There is only one universe, which has different "world-stages" at different modal indices
- **Modally Extended Objects**: Regular objects (trees, cars, people) are **modally extended entities** (i.e. perduring). Just as a person has temporal parts (childhood, adulthood), they possess **modal parts or stages** that exist at different worlds
- **Modal Tense and Reduction**: Yagisawa employs a "soft reductionism" where the "at-a-world" relation is treated as **primitive**. He uses subscripts to denote modal tense:

- **Actuality Tense:** Used for what is true at the index of the speaker (analogous to "now")
- **Mere-Possibility Tense:** Used for what is true at non-actual modal indices
- **Distinction from Lewis:** While Lewis defines worlds as **spatiotemporally isolated concrete entities**, Yagisawa views them as **metaphysical indices**. Most significantly, Yagisawa **includes impossible worlds**—indices where contradictions hold—as real points in the modal dimension

Theoretical Virtues

- **The Argument from Ways:** if we believe in possible worlds as "ways things could have been," we must, by **parity of reasoning**, accept impossible worlds as "ways things could not have been"
- **Uniform Response to Objections:** Yagisawa argues that his modally tensed language allows modal realists to solve **five major independent objections**
 - Skyrms's Epistemic Objection: Skyrms argues that if merely possible objects (like talking donkeys) are just as real and "flesh-and-blood" as actual ones, we should be able to know of their existence through the same **empirical methods** (observation, etc.) we use to know about actual things. Since we cannot "find" them empirically, Skyrms concludes they are not real
 - Yagisawa argues the objection relies on a conflation of modal tenses. Modal realists do not claim that talking donkeys exist_a but only that they exist_p
 - Our empirical methods are designed to detect things which exist_a but we can still know about them
 - **Temporal analogy:** We cannot use current empirical observation to "find" live dinosaurs today because they do not exist *now* (they exist in the *past*). Similarly, we cannot find talking donkeys because they do not exist *actually* (they exist *possibly*)
 - **Quine's Fat Man:** Posits "how many possible fat men are in the (actually vacant) doorway" is unanswerable and hence demonstrates the incoherence of possibilia
 - Reply: The question is only unanswerable because it lacks **modal-tensed specificity**. Once we disambiguate the copula ("are"), the question becomes perfectly answerable
 - Yagisawa provides several specific readings of the query. For example: "How many possible individuals who are_a fat men are_a at the doorway" → answer is zero
 - Alternatively, "How many possible individuals who are_p fat men are_p at the doorway at world w1" → answered based on census at w1

- He compares Quine's query to asking a baseball player, "How many hits did you have?" without specifying a game or season → imprecision of the untensed question
- **Peacocke's Reductio:** Christopher Peacocke argues that modal realism leads to a contradiction. He starts with four plausible premises: (A) A merely possible human is not physical; (B) Anything not physical is *necessarily* not physical; (C) A merely possible human *could* exist; and (D) Necessarily, any human is physical. Peacocke shows these are jointly inconsistent in standard modal realism
 - Reply: the argument is ambiguous because it ignores modal tense. Once specified, the contradiction disappears because **premise (B) is false**
 - Under modal tense, one way for an object to fail to be physical at a world is simply to not exist at that world → A merely possible human (like Salmon's "Noman") is_a **not a physical object** because he does not exist in the actual world. However, he is_p **a physical object** at the world where he does exist
 - Therefore, being non-physical **actually** does not entail being non-physical necessarily. Because Noman is physical in at least one world, the necessitation of his non-physicality fails, and the reductio is blocked
- **Van Inwagen on Actuality:** Van Inwagen argues that "the world we inhabit" is a **necessarily improper description**. He reasons that since each person *could* have been slightly different, each person exists in **more than one possible world**. Therefore, no unique world can be "the" world we inhabit
 - Yagisawa uses modal tense to resolve the **ambiguity** in the term "inhabit"
 - Distinguishes between inhabit_a and inhabit_p: while we inhabit_p many worlds, we inhabit_a one unique world
 - **Primitive Actuality:** To avoid the charge that "inhabita" circularly means "inhabit the actual world," Yagisawa insists that the **actuality tense is primitive and ineliminable**. Just as a temporal presentist treats "is" (presently) as a primitive concept that cannot be further defined in neutral terms, the modal realist should treat inhabit_a as a primitive indexical relation
- **Van Inwagen's Ontic Objection:** Van Inwagen reasons that if there *could have been* a million-carat diamond, then a world exists where such a diamond exists. Under Lewisian realism, this implies that a million-carat diamond exists in an **absolutely unrestricted sense**. Van Inwagen concludes that since there is no such diamond in the unrestricted sense, the theory is "outrageously false"

- Yagisawa argues that modal realists should eschew the absolutely unrestricted sense of "there is" entirely. Instead, the theory should be formulated using **modal tense**
- Distinguishes between exist_a and exist_p: Possible that there exists_a no possible million carat diamond (in actuality) but maintains that there exists_p some million carat diamond (it exists at some non-actual world index)
- **Fine-Grained Analysis:** Impossible worlds allow for the differentiation of **necessarily coextensive properties** (like "triangular" and "trilateral") and distinct impossible propositions that would otherwise be conflated into a single empty set

Criticisms

- **Exportation Principle:** Lewis and others argue that if a contradiction is true "at world w ," and w is a real concrete entity, then **a contradiction is literally true in reality**. This would make the actual world inconsistent
 - Reply: He rejects the inference that world-bound contradictions result in literal contradictions in the home language. He argues that by using **modally tensed language**, we can say that a contradiction *isp* true (at some index) without it being *isa* true (actually). Furthermore, his **soft reductionism** treats "at-a-worldness" as primitive, which some critics describe as "magical representation" that avoids the exportation problem by not requiring genuine mereological instantiation in the actual world
- **Objection to Property Identity:** Critics like Jago and Perszyk argue that if every property fails self-identity at some impossible world, then no property is identical to itself.
 - Reply: the property remains self-identical **simpliciter** (actually), and its "failure" only occurs at specifically delineated non-actual indices

PART II: Abstract Realism/Anti-realism

Sider's Overview of Abstract Realism

Linguistic Ersatzism: worlds with sets of sentences

- A possible world in which donkeys talk and fish walk would be identified with a set that includes the sentences 'Donkeys talk' and 'Fish walk, in addition to other sentences describing the rest of what occurs in this possible world
 - A talking donkey might be constructed as a set containing, among others, the formulae 'x is a donkey' and 'x talks'
- **Problem of unreduced modality:** we cannot say that a proposition is possible iff it is expressed by some sentence in some ersatz world for this would count every expressible proposition as possible. We must restrict our attention to the ersatz possible worlds, where an ersatz world is possible iff it would be possible for all members of the set to be true together
 - But unreduced modality

- Problem of unreduced modality shared by others like Plantinga's state of affairs, and Stalnaker's theory
 - Consider Plantinga's identification of possible worlds with the maximal *possible* state of affairs
 - State of affairs are abstract, necessarily existing entities. A given state of affairs may obtain or may fail to obtain (Plantinga's being an atheist does not obtain since in actuality, Plantinga is not an atheist)
 - A state of affairs S is possible iff it is (metaphysically) possible for S to obtain; say that S includes T when it is impossible for S to obtain without T obtaining; S precludes T when it is impossible for both S and T to both obtain
 - S is maximal iff for any state of affairs T either S includes T or S precludes T

Combinatorialism

- Tries to avoid circularity by giving a combinatorial definition of a possible world
- Example: possible worlds are sets of space-time points, each set representing the possibility that all and only the points in the set are occupied by matter
 - For any combination of space-time points there exists a set containing all and only those points
 - Modality is not needed to rule out impossible representations because it is intuitively plausible that any pattern of occupation of space-time points is possible
- Identifying worlds with sets of space-time points may eliminate modality from the definition of possible worlds, but reappears in the definition of 'true in'
 - There exists a talking donkey is true in a certain set S of space-time points → does not literally contain a talking donkey but in some sense represents the existence of a talking donkey
 - If we could analyse 'talking donkey' in terms of occupied points of space-time, then we could determine precisely which patterns of occupation would suffice for the existence of a talking donkey, and then we could say that it is true in S that there is a talking donkey iff S contains one of these patterns
 - But no one knows how to provide this sort of analysis of 'talking donkey' → a general analysis of modality requires a general definition of "proposition p is true in set S" for arbitrary p
 - Define: "p is true in possible world w" as "necessarily, if all and only the points in w are occupied by matter, then p is true" → circular definition but the only one possible
- Other proposals
 - Possible worlds are combinations (sets) of fundamental states of affairs

- Possible worlds are combinations of primitive atomic sentences, where a primitive atomic sentence is an atomic sentence involving only primitive vocabulary

Fictionalism

- Possible worlds as being like talk of characters in work of fiction: we speak of Holmes despite his non-existence
 - “Sherlock Holmes is a detective” is true because we really mean “According to the Doyle stories, Sherlock Holmes is a detective”
 - Rosen views talk of possible worlds as being like talk of fictional characters: a sentence, P, about possible worlds should be regarded as meaning that ‘according to the fiction of possible worlds, P’
- A sentence P, about possible worlds should be regarded as meaning that ‘according to the fiction of possible worlds, P’.
- Analysis of modality: for any modal sentence, P, let P* be the possible worlds analysis of P. For example, if P is ‘possibly, there are talking donkeys’, then P* would be ‘in some possible world there are talking donkeys’.
 - Analysis of a modal sentence P is then: according to the fiction of possible worlds, P*
- Sider: reduction of modality is doubtful since it is plausible that the locution of ‘according to’ occurring in Rosen’s analysis expresses a modal notion
 - Fiction according to Rosen’s paper is just a short summary of Lewis’s theory of possible worlds, but this falls far short of an exhaustive descriptive of a plurality of worlds. If it were possible to explicitly lay out a description D of the possible worlds, then the theorists discussed above who identified worlds with abstract entities would not have needed modal notions to define their possible worlds.
 - For instance, the linguistic ersatz could simply have defined the possible worlds as all and only the sets of sentences describing worlds in D → explicitly constraint D is difficult precisely because of the difficulty of reductively analysing modality
- Rosen’s reply: much more is true in a fiction than what a fiction explicitly states: it is presumably true in the Sherlock Holmes stories that Holmes had a liver, even though this is never explicitly mentioned. Likewise, Rosen claims, much more is true in his fiction worlds beyond what he explicitly builds into it
 - Sider: let P be any true modal sentence such that P* (P’s worlds analysis) is not explicitly entailed by Rosen’s fiction. If Rosen’s account is to be adequate, then despite his fiction not explicitly entailing P*, it must nevertheless be true according to the fiction that P*. Thus Rosen’s ‘according to’ is doing much of the world of analysing modality; accepting ‘according to’ without definition seems tantamount to accepting unreduced modality

- Fine: translates sentences about possible worlds directly into a modal language → interpret 'there is a possible individual such that p' as 'possibly, there exists an individual such that p' → goal is to reduce talk of possible worlds, not modal talk generally

Stalnaker's Abstract Realism

Key Claim: Merely possible worlds are uninstantiated properties that might have been instantiated

- Lewis's Four Theses
 - Possible worlds exist: Other possible worlds are just as real as the actual world. They may not actually exist, since to actually exist is to exist in the actual world. But they do, nevertheless, exist
 - Other possible worlds are things of the same sort of the actual world – "I and all my surroundings": They differ "not in kind, but only in what goes on at them. Our actual world is only one world among others. We call it alone actual not because it differs in kind from all the rest, but because it is the world we inhabit"
 - The indexical analysis of the adjective 'actual' is the correct analysis: 'Actual' is indexical, like "I" or "here" or "now" i.e. it depends for its reference on the circumstances of utterance, to wit, the world where the utterance is located
 - Possible worlds cannot be reduced to something more basic: it is a mistake to identify possible worlds with something more respectable such as a set of sentences of some language. We should treat possible worlds as primitives
- Against Thesis 1: Possible worlds are ways things might have been
 - Incompatible with his explanation of possible worlds as more things of the same kind as I and all my surroundings.
 - If possible worlds are ways things might have been, then the actual world ought to be the way things are rather than I and all my surroundings
 - **The way things are is a property** or a state of the world, not the world itself → the statement that the world is the way it is is true (in the sense that it satisfies the property) in a sense but not as an identity statement
- Against Thesis 2: Possible worlds are similar in kind to the actual world i.e. I and my surroundings
 - Stalnaker thinks that it is in part plausible only from equivocation between "the actual world" in the sense that is roughly captured by the paraphrase "I and all my surroundings" and which is roughly equivalent to "the way things are"
 - Deeper motivation: from thesis 3 i.e. that the essential difference between our world and the others is that we are here and not there

- Actuality as Indexical: the concept of actuality does not distinguish, from an absolute standpoint, the actual world from others
- Stalnaker: but the absolute standpoint is not a neutral one, and cannot be made distinct from the view from within any possible world
 - Instead, the **absolute standpoint just is the standpoint from the actual world** and it is part of the concept of actuality that this should be so
- Stalnaker thinks that only the semantic version of the thesis that “actual” is indexical is correct, but the metaphysical thesis that the actuality of the actual world is nothing more than a relation between it and things existing in it is false
 - We can accept the indexical analysis of personal pronouns and be a solipsist (i.e. that “I” is indexical but no one else exists)
- Stalnaker: Only the actual world alone is real because the actual world is the totality of everything there is i.e. another name for reality
 - Accepts that Possible worlds are things that might have been, where “things that might have been” is a property
 - So a possible world is a maximal way in which things could have been → unreduced modality
 - Accepts that the indexical analysis of “actual” is correct, but only in its semantic, not in its metaphysical form
 - Rejects that possible worlds are similar in kind to our actual world
- Target of Adams view: possible worlds are maximally consistent set of propositions
 - Goal: to reduce talk of possible worlds to entities which are said to be in this world
 - Stalnaker is suspicious of this: Why cannot ways things might have been elements of the actual world, as they are?
 - Two problems which Adams might be trying to solve
 - The notion of possible worlds is very obscure → possible worlds are to serve an explanatory role and so how can explanations in terms of possible worlds help us to understand anything unless we are told what possible worlds are, and told in terms which are independent of the notions which possible worlds are intended to explain?
 - Believing in possible worlds and in the indexical analysis of actuality seems to commit one to extreme realism which many believe is obviously false
 - Stalnaker thinks that his theory resolves the second because he eliminates the concrete existence of other possible worlds i.e. they are different in kind
 - But his unreductive analysis i.e. modally identifying possible worlds with things ways might have been makes the first problem about obscurity even more acute

- But an eliminative reduction of possible worlds might not be sufficient to solve the first
- Adam's view: a world-story is a maximal consistent set of propositions i.e. possible for all its members to be true together
 - But: proposition is left undefined
 - Clearly also fails to define what "possible" is → so even though it reduces talk about possible worlds, it does not make modal talk any less obscure
- Stalnaker believes that his possible worlds analysis implies the following identity and closure conditions but Adams' maximal consistent set of propositions does not (and he thinks these are plausible)
 - (I) Necessarily equivalent propositions are identical
 - (C): for every set of propositions, there is a proposition which necessarily is true if and only if every member of the set is true
 - I.e. $\{p, q\} = A$
- This is because Stalnaker identifies propositions with possible worlds while Adams identifies world-stories with propositions
 - On Stalnaker for (I): if propositions are necessarily equivalent, then they are true at exactly the same set of worlds i.e. they are just the same set of worlds and so they are identical
 - On Adams for (I): We can allow maximal consistent sets of sentences to always have (p and q) or (not p and not q) but for p to be distinct to q → Adams needs to add (I) to his theory as an additional postulate
 - On Stalnaker for (C): propositions are sets of worlds. So for any family of propositions, you can just take the **intersection** of their sets of worlds. That intersection is true exactly at the worlds where every member of the family is true. So it gives you the conjunction-like proposition required by (C)
 - On Adams for (C): even if you can talk about many propositions all being true together, it does not automatically follow that there is a **single proposition in the domain** corresponding to "all of these are true."

Rosen's Modal Fictionalism

Key Claim: we may deflate the ontology of modal realism by thinking about possible worlds as a useful fiction, under which, we may make true claims and achieve all that is good about modal realism. In other words, modal claims should be understood as claims within a fiction.

- Analogy with fiction
 - "There is a brilliant detective at 221b Baker Street" → as a straightforward existential claim, it is false

- “In the Holmes stories, there is a brilliant detective at 221b Baker Street” → perfectly true
- We can create a sentential operator of the form “In the fiction F ,...”
 - We can believe “According to F , there exists an x such that P ” without believing “there exists an x such that P ”
- Fictionalism about possible worlds: replicate the virtues of modal realism while deflating its ontology
 - Motivation: a fictionalist does not believe that there is a world with blue swans but does believe that “According to the hypothesis of a plurality of worlds, there is a world W such that at W there are blue swans”
 - Usually treat the “according to the hypothesis of a plurality of worlds” as silent
 - Leverages Lewisian concrete realism as the fiction
 - Generally: Let P be an arbitrary modal proposition. The modal realist will have ready a non-modal paraphrase of P in the language of possible worlds; call it P^* . The realist’s assertions about possible worlds are guided by explicit adherence to the scheme P iff P^* . The fictionalist’s parasitic proposal is therefore to assert every instance of the schema: P iff according to the hypothesis of the plurality of worlds, P^*
- Rosen attempts to sketch, in an incomplete way, his fiction which is basically a summary of Lewis’s concrete realism
 - Rosen’s sketch:
 - Reality consists in a plurality of universes or possible worlds
 - One of these is the one we inhabit i.e. spatiotemporal system of which we are parts
 - Others are of roughly the same kind
 - Each universe is isolated from each other
 - Totality of universes is closed under a principle of recombination
 - No arbitrary limits on the plenitude of universes
 - Add: The universes are possible worlds, our universes and its parts are actual individuals; the others are merely possible
 - Rosen admits that these original postulates are not quite enough for the fictionalist’s purposes: they say rather little about what sorts of objects the various universes contain → nothing guarantees that some of the universes contain people or clouds or stars
 - Proposal: supplement with an encyclopedia: a list of the non-modal truths about the intrinsic character of the universe
 - Specifies, for instance, that our universe contains objects which are rather like ordinary kangaroos, but for the lack of a tail. So, by recombination, other universes contain

free-standing tailless kangaroos, duplicates of these residents of our world

- The fictionalist's fiction is the sketch of modal realism together with the complete catalogue of non-modal intrinsic truths about our world
- Fictionalist theory of possibility
 - P iff according to PW, P*
 - According to PW, there is a universe containing blue swans
 - In central cases, the realist and the fictionalist will concur in their theoretically informed modal judgements
 - Whenever the realist believes that his paraphrase of a modal claim is true, the fictionalist should have the same view of his paraphrase
- Advantages:
 - No revisionary ontological consequences: no need to believe in possibilia but can believe everything the fictionalist says about the truth conditions for modal statements
 - Fictionalism does make an ontological commitment to entities when talking about stories, theories and other representations
- Epistemology: whether fictionalism is compatible with the view that our usual methods for forming modal beliefs are generally a good guide to modal truth
 - Lewis: our everyday modal opinions are, in large measure, consequences of a principle of recombination
 - Basically the role of imagination in modal reasoning → we construct representations of situations in accordance with our empirical knowledge, the principles of recombination, non-arbitrariness and so on and if this route is aptly described by the statement P, then "Possibly P" is true
 - We must explain why this imaginative method is a reliable way of finding the truth → realist has no answer other than we are lucky
 - Realist answer: (i) modal truths are truths about a domain of universes; (ii) the principles which guide our imagination are true of that domain; so (iii) by and large, when we imagine in accordance with these principles, the state of affairs we image are realised somewhere along the universes
 - Rosen: reject (ii) as a striking conjecture → is it just a coincidence that the principles which guide our imagination truly describe a domain of objects with which human beings had absolutely no contact when those principles were being shaped, presumably by a perfect natural evolutionary process
 - Why is it so lucky that we have been given the right imaginative dispositions
 - No analogous problem arises for the fictionalist

- If the realist is right in suggesting that we are guided in the imaginative construction of possibilities by principles like the postulates of PW, then when we engage in imaginative experiments, the least we discover is what is true according to PW
 - For the fictionalist, that is enough → modal facts just are facts of this kind (facts that emerge from our imagination)
 - No special mystery as to why we should trust our imaginations as a guide to modal truth: it is after all, a fiction we use our imagination to construct
- Incompleteness problem: marginal cases which the fictionalist and realist must disagree about what the modal truths are
 - Example of a modal claim which the realist thinks has a determinate truth value of which the fictionalist is ignorant → ignorance must not be due to ignorance about empirical facts and must be robust i.e. surviving an arbitrary extension of our scientific and historical knowledge
 - Not obvious such statements exist
 - Lewis: claims such as how large the universe might have been fit into this category
 - Example: let x be a cardinal number larger than the number of space-time regions in our universe and consider
 - There might have been x non-overlapping physical objects
 - This, for the realist, is a claim about the size of our largest universes: there is a universe containing x non-overlapping physical objects
 - But there must be an upper bound x^* to the number of non-overlapping objects that inhabit the most populous worlds but we do not know what this upper bound is even if we were fully informed of the empirical character of our universe
 - But the modal realist can still say this statement has a definite truth value – we just don't know what it is
 - For the fictionalist, the story is simply silent on this point because the list of postulates and a complete account of the non-modal intrinsic nature of the world, we might still be in no position to affirm that according to PW, there are universes containing x objects
 - Possible reply: simply say such indeterminate sentences are false
 - Why: According to PW, there is a universe with x non-overlapping physical objects → says that the realist theory settles whether there are worlds of a certain size but it clearly does not
 - But problem: The negation then must be taken as true → but the realist hasn't settled this either so it must be false if the first is false

- Rosen's solution: lack a truth value
 - In general, when the paraphrase P^* of a modal claim P is not determinately settled as true or false by the theory PW , the modal claim P is to lack a truth value
 - Stipulation: ordinary logical connectives no longer truth functional because the disjunction of a statement and its negation is always true even if they lack a truth value
- Recalcitrant problem: incompleteness in fictions do not usually mean truth-gaps
 - Example: Holmes stories are silent about Moriarty's blood type but when someone asserts "In the Holmes stories, Moriarty has blood type A+", we would naturally suppose that he had said something false, not something lacking in truth value
- Reduction
 - A central feature of the fictionalist's strategy is the use of the prefix "**According to the story PW ...**" (where PW is the fiction of a plurality of worlds). This operator is treated as an intuitively understood, **undefined primitive**. However, this leads to two main objections:
 - **Failure of Reductive Analysis:** Unlike the modal realist, who claims to provide a full reduction of modal terms into non-modal terms, the fictionalist's analysis contains an **unreduced modal component**. If one tries to explain what "According to PW , P " means, one typically uses modal language, such as "If PW were true, then P would be true"
 - **Obscurity of the Primitive:** Critics argue that the fictionalist's primitive is actually more obscure than the modal notions it is intended to explain
 - Reply 1: Rosen argues that even if the reduction isn't "eliminative," reducing a wide variety of modal notions to a single primitive is still a **non-trivial analytic advance**
 - Not true even of modal realists: many recognisably modal judgements cannot be expressed in a language whose modal vocabulary is restricted to boxes and diamonds
 - Reply 2: If the fictionalist's prefix is a modal operator, then it is by no means obvious that the realist has given a fully reductive treatment of modality; for it remains unclear how the realist proposes to eliminate occurrences of this very locution.
 - While the realist claims to be able to analyse all modal locutions in non-modal terms—a "full reduction"—the fictionalist argues that the realist lacks a reductive account for the very class of story prefixes that includes "**According to the story PW ...**"

- The realist does have a powerful strategy for eliminating a large class of story prefixes used in ordinary narrative fiction. Under this analysis, the content of a story *S* is identified with a specific set of possible worlds (*Cs*), and a sentence *P* is "true according to *S*" if *P* is true in all the worlds in that set
 - For example, the content of a Sherlock Holmes story can be reduced to the set of worlds where the text of that story is a record of known fact. However, the sources argue that this strategy **cannot generalize to metaphysical stories** that represent the totality of modal reality
- Failure from metaphysical stories: The central problem is that a story about the **totality of worlds** must be either necessarily true or necessarily false from the realist's perspective. If the content of such a story is represented as a set of worlds, it would have to be either the set of **all possible worlds** or the **null set**. Neither of these options captures the determinate content of a specific metaphysical speculation
- **Example:** Consider *This Lonely World*, where the reader is invited to imagine that **only our universe exists** and everything happens necessarily
 - The realist cannot represent this story's content as a set containing only the actual world, because that would just be a list of actual facts, missing the story's claim that *nothing else is possible*
 - The realist has no reason to believe in "lonely worlds" (worlds where no other worlds are possible), making it difficult to find a set of worlds that accurately represents the story's "grim implications"
- Problem: there is just something unsatisfying about the fictionalist's choice of primitive
 - Gives rise to truth value gaps and does not feel primitive
 - Truth relative to a story sounds like the sort of thing one ought to be able to explain, in stark contrast to more plausible primitives like negation
 - "True according to PW" is simply too obscure
- Objection from Concern
 - Humphrey Objection but modified: why would Humphrey care that there is a story the content of which says that he would have won i.e. according to PW, there is a world in which someone rather like him – his counterpart – does win

- Could he coherently feel regret → thinking about what happens in fiction is not centrally important to us
 - To regret having acted in a certain way, one must think that one had acted differently, things might have been better
- Parallel Lewis's reply: a lucid fictionalist may come to think that truth in the fiction PW is important to him in a way he had not previously imagined
 - The fact that things would have gone better if he had only acted differently just is the fact that according to PW, worlds most like the actual world in which his counterpart act differently are worlds where things go better
 - But a psychological question
- Timid Fictionalism
 - Rosen: the viability of fictionalism depends on one's philosophical goals.
 - Timid Fictionalism: for those who only want a "license to move back and forth" between modal talk and possible worlds talk without claiming to explain the fundamental nature of modal truth.
 - Transitions guided by biconditionals: P iff according to PW, P^*
 - Need not treat these biconditionals as providing genuine analyses
 - For this "timid" version, many of the deeper philosophical objections—including the argument from concern—simply do not arise, as the theorist does not claim that modal facts are identical to fictional facts. Ultimately, fictionalism remains an attractive option for those who wish to use the powerful framework of possible worlds while maintaining a sensible one-world ontology

Melia's Linguistic Ersatzism

- Identifies possible worlds with sets of sentences
 - Man problem: generating enough books (set of sentences) to represent all the possibilities we pre-theoretically wish to accept
- Still realism: worlds do not depend upon facts about language users because books (sets of sentences) can be identified with set-theoretic constructions of particulars and universals which would exist without language users
- But: without language users, strings of symbols would not represent anything and so the worlds themselves may exist independently but what is the case according to these worlds do not (sort of like Churchill-Ant example)
 - Reply: World making language of the linguistic theorist is not one we need speak or right down and can be far removed from the languages we use
 - Simply need to adequately define the phrase "true at" in for all possible P , there are many entities w such that P is *true at* w

- Wants: given definition of “true at”, the fact that donkeys talk is true at w is a fact that holds entirely independently of the existence or conventions of language users
- World-making
 - Language: there should be enough sentences to represent all the possible ways the world could be, or the resultant theory would fail to recapture the equivalence between P ’s being possible and the existence of a world at which P is true
 - Nice to have names for every object and predicates for every property or at least a set of predicates rich enough to express every property
 - Lagadonian solution: let each particular name itself and each property or relation be its own predicate
 - Let atomic sentences be n -tuples of the form $\langle R, a, \dots, b \rangle$ where R is an $(n-1)$ -place relation and $a, \dots, b$ are $n-1$ particulars $\rightarrow a \dots b$ stand in the relation R
 - Note: no need to name necessarily existing objects such as numbers and sets and only need fundamental properties
 - Possible worlds: maximally consistent atomic sentences
 - Accept the non-circular definitions of necessity and possibility $\rightarrow$ through the use of entailment (below) and maximal consistency
 - Ontology: similar to combinatorialism
 - Safe and sane: all its postulates are actually existing particulars, actually instantiated properties, and set-theoretic constructions thereof
 - Representation: no mystery $\rightarrow$ recombinations are to be understood as sentences whose content can be spelled out as follows
 - For any particular a , and every property F , the atomic sentence “ a has F ” is true at w iff $\langle F, a \rangle$ in w
 - For any particulars $a, \dots, b$, and every relation R , the atomic sentence “ $a \dots b$ are related by relation R ” is true at w iff $\langle R, a, \dots, b \rangle$ in w
 - For any non-modal sentence S , S is true at w iff the atomic sentence trues at w entail S
- Immediate limitations
 - No ontology of possibilia $\rightarrow$ cannot give a translation of “There could have been more things than there actually are”
 - Possible reply: possibilia can be taken to be maximal sets of one-place predicates, each predicate describing a complete way a possible object could be
 - A particular possible object exists at a world precisely when its existential closure (binding all the free variables with “exists an x such that”) at the world w

- Alien particulars and properties: sentence of the linguistic theorist's world-making language only contain names and predicates for actual particulars and predicates for actually instantiated properties
 - Intuitive appeal of alien properties: there are worlds with actually uninstantiated properties
 - Prima facie solution (for particulars): can generate a sentence of the form "there exists an x such that $(x \neq a) \text{ and } (x \neq b) \dots (x \neq c)$ " where $a, \dots, c$ are the list of particulars actually existing
 - Don't need names because we didn't need names for numbers of pure sets
- Haecceitism: there are distinct worlds agreeing upon which first-order roles are occupied but disagreeing over which individuals play which roles (there can be qualitatively identical but distinct worlds)
 - Consider
 - W1: a and b are F , c is G
 - W2: a and c are F , b is G
 - First-order roles: Suppose that sentence S , a (possibly infinite) conjunction, describes in perfect detail the qualitative character of some possible world. Replacing every occurrence of some name with a free variable, and then deleting each conjunction not containing that variable, results in a conjunction of open formulas that describes a role a particular may occupy
 - Plausibility of distinct worlds that differ only over which objects play which roles → problem is that they are qualitatively identical
 - Imagine a world containing a solitary cylinder balanced on a plane. Suppose that the situation is perfectly symmetric: all parts of the cylinder are made of the same material and each atomic part of the cylinder has the same intrinsic properties as any other and the object is a perfect cylinder. Similarly, the plane on which the cylinder is resting is completely homogenous: all points of the plane share the same intrinsic properties. After a certain time t , the rod falls over in a random direction. Intuitively, there are many different ways the rod could fall. We might like to spell this out by saying there are many different worlds where the rod falls in different directions. However, all these worlds are qualitatively indiscernible: they differ only over which points of the plane the cylinder ends up falling towards
 - Basically no genuine difference (no north/south etc)
 - Second Order Haecceitism
 - Consider a sentence S : replacing every occurrence of some predicate with a free variable and then deleting each conjunct not containing that variable results in a conjunction of open formulas that expresses a second-order role

- I.e. w_1 : a and b are F, c is G vs w_2 : a and b are G vs c is F $\rightarrow$ agree which second order roles are occupied but disagree as to which properties occupy these roles
 - Second order haecceitism not as widely debated but more more popular (Lewis subscribes to it)
 - Weak nomological essentialism: a property may occupy its nomological role vacuously (F may be said to play this role even if there are no objects that instantiate G for it to interact with)
 - Plausible: Consider w and w' such that both only contain object a but a instantiates different properties at the two worlds. The second order role occupied by the two worlds is expressed by Xa . Thus at both w and w' , precisely the same second order role is being occupied: being instantiated by a. But a world containing one particle that instantiates the property being charged is surely different from a world containing a particle that instantiates the property having mass
 - These are distinct worlds that differ only over which properties occupy which roles
- Alien properties: linguistic theory conflates distinct possibilities, for a scheme that uses existential sentences to generate worlds containing alien properties is unable to generate worlds that differ only over which alien properties occupy which roles
 - They can only represent worlds with alien properties by saying that there are so-and-so roles that are occupied by alien properties but do not say which properties occupy these roles
 - We can see this by considering the schematic form “there exists an x such that $(x \text{ not} = a) \text{ and } (x \text{ not} = b) \dots (x \text{ not} = c)$ ” an drawing an analogy to properties
 - Example: Consider two worlds V and V' . At V , we have a and b are Q and c is Z while at V' , we have a and b are Q while c is Z. These are clearly distinct worlds since we have 2Qs in one world and 2Zs in the other. But the linguistic theorist can only represent the world V with (in SOL)
 - There exists property X such that there exists a property Y such that (X is not F and X is not G and ... and Y is not F and Y is not G ... and X is not Y and Xa and Xb and Yc)
 - But this also represents V' !
- Solution: linguistic theorist need not conflate the possibilities because having the predicates of alien properties is only sufficient but not necessary for distinguishing these worlds
 - Consider V and V' which are supposedly conflated. V is distinguished from V' by

- The property that a and b instantiate at V is not identical to the property that a and b instantiate at V' but is identical to the property that c instantiates at V'
 - The property that c instantiates at V' is not identical to the property that c instantiates at V but is identical to the property that a and b instantiate at V
- But these two facts involve merely the transworld identity or non-identity of properties: all the linguistic theorist needs to say is which properties playing which roles at V are identical or distinct from which properties playing which roles at V'
 - No need to say which properties play which roles
- Need to enrich world-making language to represent transworld identity: just as the identity of actually existing atomic entities across worlds was represented by the same particular entity appearing in different sets of sentences, so shall the identity of alien entities across worlds be represented
 - Because alien properties and particulars are not part of our ontology, need to use pseudo-predicates/names
 - For names, Fc_j where c_j does not denote anything but interpreted as existentially i.e. Fc_j is true iff there is some object that is F
 - $\{Fc_j, Hc_j\}$ and $\{Fc_j, Gc_j\}$ have the same object satisfying H in the first and G in the second $\rightarrow$ use of c_j across both represents sameness
 - Same for properties: use pseudo-predicates such as C_j and if C_j is in w and C_k is in w' we interpret it as there being some property X relating a,...,b in w and some property Y relating a,...,b in w' but $X \neq Y$
- Objections
 - Lewis: These linguistic representations do not represent the many possibilities unambiguously (since existentially) i.e. no one-one correspondence
 - Melia: not true that it is ambiguous because the sentences have a determinate meaning (from its own formulation) even though it does not represent possible worlds 1-1
 - Obviously a stupid argument
 - Melia:
 - Cannot see how different ways in which the existential sentence could come true are represented
 - Cannot see how the construction manages to generate possible worlds where an alien entity plays role p at one world and plays role q at another for each representation says exactly the same thing

- Pairing each representation with a number merely results in a set of sentences that all say the same thing
 - Example: plays the role p at the world [d and p(C_j) and q(C_k)] plays the role q at the world [d and p(C_k) and q(C_j)]
→ even if the names refer to distinct properties, the construction still does not explain how we get **genuinely distinct possible worlds** rather than just different representational assignments.
- Melia's proposal
 - Alien property that plays the role p at the world [d and p(C_j) and q(C_k)] plays the role q at the world [d and p(C_k) and q(C_j)]
 - The two worlds [d and p(C_k)] and [d and p(C_j)] represent different ways in which the sentence "there exists an X[d and p(X)]" could be true because the alien properties that are instantiated according to the two sentences are different
 - Different realisations of the same sentence
 - Lewis: each such world contains a pseudo-predicate for an alien property represents not only a possible world, but ambiguously represents many
 - Even if not ambiguous, it does represent many → so the descriptions of the worlds are not complete
 - Melia: no reason to believe why possible worlds should be complete
 - Stalnaker: nothing in the formalism of possible world semantics, or in the intuitive conception of a way things might be, or a possible state of the world, excludes an interpretation in which possible worlds are alternative states of some limited subject matter
 - Just because a pseudo-predicate is incomplete does not mean they fail to represent whether or not the alien entity that exists according to one such world is identical or different from the alien entity that exists according to another such world

Fine's Essentialism

Key Claim: Essence cannot be understood in terms of necessity, and instead is metaphysically prior to necessity. So, necessity and modal claims are grounded in terms of essences. Essence, for Fine, is a finer mesh than modal notions

- Significance of essences: may be used to characterise what the subject, or at least part of it is about and is a property which bears, in the metaphysically significant sense of the phrase, on what the object is
 - The metaphysician may want to say that a person is essentially a person or that having a body is not essential to a person
 - Use in definitions of metaphysical concepts: an essential being is one whose essence includes its own existence

- Concepts of substance and ontological dependence
 - A substance: something whose essence does not preclude it from existing on its own
 - One object depends upon another if its essence prevents it from existing without other objects
- Two lines of progress
 - Defining essence in modal terms
 - Essence resides in the “real” or objectual cases of definition
- Modal Definition of Essence (which Fine rejects): through de re modal attribution
 - Formulate essentialist claims via quantified modal logic → upshot: nothing special about the modal character of these essentialist claims beyond their being de re
 - Simplest Form: Takes an object to have a property essentially just in case it is necessary that the object has the property i.e. x is an F in all possible worlds
 - Variant I: makes necessary possession conditional on existence i.e. an object is taken to have a property essentially just in case it is necessary that the object has the property if it exists
 - Variant II: makes necessary possession conditional upon identity i.e. an object is taken to have a property essentially just in case it is necessary that the object has the property if it is identical to that very object
- Fine: essence cannot be understood in modal terms or even to be regarded as extensionally equivalent to a modal notion; instead, we should use the traditional assimilation of essence to definition, even though it does not provide us with reductive analysis of essence
 - Immediate suspicion of modal accounts: might be a misreading of our informal way of saying an object essentially has a certain property “an object must have that property if it is to be the object that it is” → goal is how to understand the qualifying phrase “if it is to be the object that it is”
 - Under the condition of existence: this is expressed in terms of identity but Fine questions why is the existence of the object expressed so perversely in this way
 - Under the condition of identity: the phrase conveys a vacuous condition – then why is the qualification of identity made in the first place
- Fine: modal accounts are insufficient (but necessary) for an object to essentially have some property
 - Accept: if an object essentially has F , then it necessarily has F
 - Reject: if an object necessarily has F , then it essentially has F
 - Example: Socrates and the set whose sole member is Socrates. If Socrates exists, then he belongs to the singleton $\{\text{Socrates}\}$ for necessarily, the

singleton exists if Socrates exists, and necessarily, Socrates belongs to the singleton {Socrates} if both Socrates and the singleton exist. Then according to the modal criterion, Socrates essentially belongs to the singleton Socrates

- Reject this implication that it is essential for Socrates to belong to the singleton Socrates
- Another example: Socrates and the Eiffel Tower are necessarily distinct (because their natures are unconnected). But it is not essential to Socrates that he be distinct from the Tower; for there is nothing in his nature which connects him in any special way to it
- Modal account subject to further difficulties: consider any necessary truth such as “ $1+1=2$ ”. It is necessarily the case that this truth should hold if Socrates exists but it is no part of Socrates’ essence that “ $1+1=2$ ”
- Does not apply just to the conditional accounts but also to the categorical accounts i.e. “necessarily x has F”
 - Modification: use necessary existents such as “2” and the singleton “{2}”
 - Another Modification: make attribution of essential property conditional on existence
 - Instead of asking whether Socrates is essentially a member of the singleton Socrates, we ask whether he is essentially a member of the singleton if he exists
- Problem of necessary existence (for conditional accounts)
 - It is necessarily the case that Socrates exists if Socrates exists → but it is not essential to Socrates that he exists
 - It is necessarily the case that Socrates’ parents exist if Socrates exists → but it is not essential to Socrates that his parents exist
- Possible solutions:
 - Make the demand that an essential property not be an essential property of every object whatsoever
 - Then the examples with necessary truths do not hold (since they necessarily hold with every object)
 - But doesn’t defuse everything
 - Condition of relevance to the object
 - Fine: but we do not want to impose a general ban on “improper” properties being essential; we might be happy to say, for instance, that it is essential to the null set that there be sets
 - Possible to construct counterexamples with “proper” properties
 - Also circular: how to give an account of relevance without presupposing essence in question
- Basically: Fine thinks there is no modal account of essence that is satisfactory and this is particularly important for our understanding of the metaphysics of identity

- Why does a modal account go so badly: necessity in an object necessarily having an essential property is not necessity simpliciter → it is true in virtue of the identity of the object in questions and necessity has its source in those objects which are the subject of the underlying essentialist claim
- Thus, different essentially induced truths may have their source in the identities of different objects → an induced truth which concerns various objects may have its source in the nature of some of these objects but not of others
 - Socrates being a man having its source in the identity of Socrates
 - 2 being a number having its source in the identity of 2
- Metaphysical necessity is insensitive to source: all objects are treated equally as possible grounds of necessary truth
- Fine: Metaphysical necessity is a special case of essence
 - Metaphysically necessary truths can be identified with the propositions which are true in virtue of the nature of all objects
 - Logical necessities can be taken to be propositions which are true in virtue of the nature of all logical concepts
- Essence and real definition
 - Analogy with analyticity: A sentence is analytic if it is true in virtue of the meanings of its terms. For example, "all bachelors are unmarried men" is true in virtue of the meaning of 'bachelor' but not in virtue of the meanings of "unmarried" and "man"
 - Relativised form of necessity: Necessary truth may be true in virtue of the identity of certain objects as opposed to others
 - Essential Truth: Similarly, a proposition is essentially true if it is true in virtue of the identity (essence) of the objects involved
- Relationship between defining a term and giving the essence of an object
 - One results in a sentence which is true in virtue of the meaning of the term and the other results in a proposition which is true in virtue of the identity of the object
 - Fine thinks they are at the same: the only satisfactory answer to an appropriate specification of meaning is that the specification should make clear what the meaning (essentially) is; it should provide us, that is to say, with some account of the meaning's essence
- Fine challenges the contemporary prejudice that only words or concepts can be defined,. He argues that the activities of specifying the meaning of a word and stating what an object is are **essentially the same**; both have an equal right to be regarded as forms of definition
 - He suggests that if it is meaningful to define the concept 'bachelor' as 'unmarried man,' it is equally meaningful to define a particular set in terms of its members or a molecule of water in terms of its atomic constituents

- In Fine's view, stating an object's essence is providing its real definition—stating what the object is

Part III: Elimitivism

Quinean Skepticism

Key Claim: modal notions are, in some sense, not required in our philosophical and scientific theorising. Grave problems in interpreting intensional logics especially quantified modal systems

- Intensionality and compositionality
 - Quine: preferred extensional systems over intensional systems
 - Extensional systems: First order logic → different parts of this language have a certain kind of entity associated with them and these entities are called extensions. Extensions of predicates are the set of things that satisfy the predicate, extensions of sentences are truth values etc
 - Extensionality: let p and q be formulas or terms of our language and suppose that p contains q . A language is extensional if, whenever q^* has the same extension as q and whenever p^* is the result of replacing all occurrences of q with q^* , then the extension of p is equal to that of p^*
 - Intensionality: a system is intensional just when it fails to be extensional
 - Let p be the modal sentence "Necessarily, x " and suppose that y has the same truth value i.e. extension as x . p^* will be "Necessarily, y " but this might have a different truth value → because x and y only have the same extension at the world w (and may have different truth values elsewhere)
 - Melia: possible to ditch QML and expressed all our modal theses in an extensional first order language that quantified over possible worlds and possibilities
 - Reason to prefer extensionality: compositionality
 - The extension of a whole formula is a function of/depends on the extension of the parts of the formula
 - Compositional semantics: If we want to explain our ability to use and understand a potentially infinite number of sentences, we must suppose that the meanings of complex parts of language depend in a rule-based way on the meaning of simpler parts of language
 - Reply: all that is necessary for compositional semantics is that the meaning of complex strings of symbols depend upon meanings of its parts. No argument for the truth value of a sentence to be a function of the extensions of its parts
 - Extensions introduced merely be stipulation
 - Possible to do this with modal languages → identification of truth with truth-in-the-intended model

- Melia: just because possible worlds never appear in the list of extensions does not indict modal logic but could demonstrate the poverty of merely considering extensions
 - Not radical that a theory of meaning will need entities not found on the list of extensions → Frege: names have both a reference and a sense
 - Clearly, it becomes an ontological question about considering these extra-extensional objects (beyond truth-values and referents etc)
- So: no real principled distinction between extensional and intensional systems → issue is now one of ontology
 - Possible worlds semantics can meet the challenge of compositional semantics because they do have such semantics
 - Merely ontological worries, not a logical one
- Referential Opacity: by substituting different terms referring to the same individual in a modal sentence, the truth value of the whole sentence is affected
 - Extensional contexts: if “ $a=b$ ” is a true identity statement, then either one of the terms a and b may be substituted for each other in any true sentence S containing one of those terms
 - If “Joe is the tallest man in class” is true and the identity statement “Joe is the sole occupier of Terrace number 7” then the statement “The sole occupier of terrace number 7 is the tallest man in class” is true
 - Quine: extensional systems are referentially transparent
 - Paradigmatic example for failure for modal logic
 - Consider “ $8 = \text{the number of planets}$ ” is a true identity statement
 - Consider “Necessarily, 8 is greater than 7” is also a true modal statement
 - But “Necessarily, the number of planets is greater than 7” is false
 - Melia: QML does not have the linguistic resources capable of forming terms such as “the tallest man” → instead, it deals with sentences containing definite descriptions by paraphrasing them per Russell
 - The tallest man is tall becomes “For any x , if x is a man taller than any other, and x is the only man taller than any other, then x is tall”
 - This paraphrase makes it difficult to see why substitutivity fails
 - But suppose that a decent modal logic ought to have the linguistic resources to formulate descriptive referring terms
 - Some: referential opacity a cause for concern → “if identity does not mean universal interchangeability, then I do not really understand identity at all”
 - Reply: But all this shows is that a form of argument valid for predicate logic turns out not to be valid in quantified modal logic

- All that follows from this is that we must rewrite the inferential rules for QML i.e. restricting the principle of substitutivity
- Quine: indiscernibility of the identicals → if a and b are identical they should have the same properties and so we should be referentially transparent
 - Melia: Not clear how this metaphysical principle supports the logical principle that our systems ought to be referentially transparent and Quine simply assumes that referential transparency is the formal implementation of the indiscernibility of the identical
- Most important objection: in modal logic, we should expect referential opacity
 - Every should agree that there are terms to have their reference relative to a particular context e.g. indexicals such as “here” or “prime minister”
 - Even in such contexts, we should not expect referential transparency to hold
 - “Tony Blair = The Prime Minister” is true and “At some point in the future, the prime minister will be a woman” is true but “At some point in the future, Tony Blair will be a woman” is false
 - Even though the two objects refer to the same thing now, might not in the future
 - Possible worlds theorists think that referential opacity occurs for precisely analogous reasons: depends on which world the proposition p is in
 - At the actual world, Venus and the Morning star refer to the same object but there is no reason to expect that in other worlds, they refer to the same objects
 - That “a=b” is only to be told in the actual world, these terms are co-referential → indiscernibility of the identicals is consistent with referential opacity in modal contexts
- Quine’s objection to systems without definite descriptions (such as QML)
 - Different grades of modal involvement → the higher the grade, the more unhappy Quine is
 - First Grade: Necessity is understood as a predicate of sentences i.e. a property of sentences
 - “Necessarily, $2+2 = 4$ ” is to give a property to the sentence “ $2+2=4$ ”
 - Substitutivity immediately solved because there is no inference from “Necessarily, 9 is greater than 7” and “9 equals the number of planets” to “Necessarily, the number of planets is greater than 7” because the term 9 doesn’t appear referentially in the two modal sentences
 - Analogous invalid inference

- “9 is greater than 7” contains 5 words
 - 9 is the number of planets
 - C: “The number of planets is greater than 7” contains five words
- Different sentences
- Second grade: “necessarily” is taken to be an operator that attaches to closed formulas
 - Given a sentence P, “Necessarily P” is also a sentence → “necessarily, $2+2=4$ ” is to be understood like “It is true that $2+2=4$ ”
- Third grade: “Necessarily” can be used as an operator on both closed and open sentences
 - “Necessarily x is greater than 7” can be quantified over “There is some x such that necessarily, x is greater than 7” → Quine finds this absolutely intolerable that the quantifier can bind a free variable that occurs within the scope of a modal operator → commits us to nonsense or untenable essentialism
- We can accept the second grade even if we give in to Quine’s argument
- Quine’s objection: whether or not necessity attaches to a particular object is a matter of the way in which the object is specified
 - “There exists an x such that necessarily, $x > 7$ ” is true if we specify the object using the name “9” but false if we specify the object using the description “the number of planets”
 - Not the same as referential opacity: cannot explain away by saying that the two terms refer to different objects in different worlds because the “necessity” in “Necessarily, x is greater than 7” cannot be read as “is true at all worlds” since it contains a free variable → doesn’t even have a truth value at the actual world
- Reply: de re/de dicto distinction
 - If Quine’s question is interpreted as a de re question which object, in and of itself, has the property of being greater than 7 necessarily, then all hands agree that if this is true of the number 9, then it is equally true of the number of planets for these are one and the same
 - Quine is wrong to suggest that judgements on the de re proposition will change depending on how the object is specified
 - Tempting because we might hear the proposition as a de dicto one: “Necessarily, the number of planets is greater than 7” is false → but this is a different proposition
- Quine’s essentialist charge: unclear to Melia what these contentious essentialist consequences are supposed to be

- Can avoid that in and of itself, an object could have a property essentially by simply saying we can reject de re claims as being false
- The preceding de re/de dicto argument was merely to say that it is not true that “There exists an x such that necessarily, $x > 7$ ” changes based on how we specify x

Modalism vs Anti-Modalism

Key Claim: Modalism is the view that we can just give an account of modality via modal notions. Melia rejects modalism and argues that we can explain modal notions even if we have used the concepts of possibility and necessity in explaining what a possible world is

- Modalism: Analysis of modal statements only through modal operators and not through quantification over any kind of entity
 - Taking necessarily at its face value
 - Can be a modal realist: hold that there are objective modal facts without having to enter the difficult debate over the nature of possible worlds
 - Rests on Two Claims: (1) that any modal claim we wish to make is expressible in a language which neither quantifies over possible worlds or possible objects; (2) that a modalist is justified in believing that possible world semantics does give a reliable guide to the logical and semantic properties of modal languages
- Anti-Modalism: Treat modal statements as involving quantification over worlds
- Against Claim (1): that any modal claim we wish to make is expressible in a language which does not quantify over anything
 - Compare a modal language with a two-sorted extensional one that quantifies over worlds → goal is to show that there are sentences expressible in the latter not expressible in the former
 - First Order Modal Language L contains one-place predicate ‘E’ (for exists), two place predicate =, operators box and diamond, a set of constants, predicates and symbols of an ordinary FOL
 - Model theoretic semantics for this language → two operators box and diamond treated as universal and existential quantifiers over possible worlds
 - Worlds only appear in the semantics of the theory of the language, not necessarily in the ontology of the theory (do not treat worlds as denoting actual things)
 - Forbes adds the actual operator @ to express sentences like “There could have been things which do not actually exist”
 - Two sorted extensional language which basically adds a second dimension for worlds (and first dimension is for variables)
 - I.e. E is a two place predicate for variables and worlds like “ Exw ”

- Suspicion: if there exists a two-sorted extensional language $L_{\{EX\}}$ identical to L^* and for any sentence $L_{\{EX\}}$ there is a sentence L' which is its translation and which is structurally very similar to the sentence of $L_{\{EX\}}$ then $L_{\{EX\}}$ makes it likely that L' does indeed make use of possible worlds
- Basically, Forbes' language is quantifying over worlds – because of the structural similarities between $L_{\{EX\}}$ and L'
- Against Claim (2): possible world semantics provides a reliable guide to the logical and semantic properties of modal languages
 - Forbes does not believe that the model theory (i.e. quantifying over possible worlds) correctly represents the state of affairs that make modal sentences true
 - Justifies his semantics by claiming to see a meaning equivalence between “Possibly, P” and “There is a possibility that P” and that the latter gets its meaning from the former
 - But this just introduces a new kind of abstract object called possibilities
 - Forbes: in how discerning genuine quantification structure in “there is a possibility that P” after having maintained that there is no such structure in its ordinary meaning, we are justified in thinking that the new interpretations yield a reliable guide to the logico-semantic properties of the original modal language because of the **conception of a possibility as something that makes a sentence true**
 - If a possibility that would make such-and-such sentences true would also not make another sentence true, then an argument with exactly those sentences as premises and conclusion would be invalid; while if a specific modal argument-form is invalid, there must be a possibility that makes the premises true and the conclusion false. Incomplete possibilities basically give the same semantics as complete worlds
 - Melia's objection
 - The model theory is justified in giving a guide to the logical and semantic properties of sentences which only use box and diamond but Forbes language L uses more than them i.e. @
 - No way to have something like @ in modal theory: suppose there is a meaning equivalence between “actually P” and “at the actual P”; then would be impossible that the model theory should make false/true a sentence by which its intuitive meaning we judge to be true/false since Forbes thinks that model theory get its meaning from intuitive concepts
 - But impossible for “actually”: “If P is actually the case then necessarily P is actually the case” is intuitively false but comes out true on the model theory

- Forbes argument works only if it justifies us in taking some aspects in model theory seriously, but then it needs to suppose that model theory represents, in some serious way, reality i.e. quantifying over possible worlds
 - Possibilities alone are not the things that make modal statements true
 - Diamond Diamond P is not made true by the existence of a possibility possibly P but rather is true at a possibility iff there is some possibility accessible to it at which there is the accessible possibility that $P \rightarrow$ requires understanding the range of entities, possibilities, between which certain relations hold
- Explaining modal concepts through quantification over possible worlds
 - Problem: how do we provide a genuine analysis when we are told that “Possibly P” when possible worlds are defined using unreduced modality
 - Quantifying over Possible worlds as explaining modal concepts: (1) counterfactuals, (2) helps us see correct inferences that hold between statements using modal operators
 - Example of counterfactuals: possible worlds analysis of these statements is explanatory in the sense that it improves our understanding of such statements, and is also explanatory in the sense that someone who did not understand the counterfactual conditional would be enlightened when told ‘If P had been the case then Q would have been the case’ means “At most similar world where P is true, Q is true” even if world is identified with some concept involving unreduced modality
 - For (2), Forbes thought that we grasp inferential properties of box and diamond and perceive the validity of simple modal arguments forms such as $\Diamond (P \ \& \ Q) \mid - \Diamond P'$ without having ever thought of the quantificational interpretations of operators
 - Melia: untrue for iterated modalities: intuitions fail to tell us whether “Possibly, possibly P” implies “Possibly P” etc (i.e. the axioms of different relational frames T, B, S4 or S5)
 - Instead, our notions of possible worlds help us choose which axioms
 - Example: Plantinga and maximally consistent states of affairs. He argues that “among the necessary beings, furthermore, are states of affairs and hence possible worlds themselves” as well as that maximally consistent sets of states of affairs do not vary from world to world $\rightarrow$ so the correct analysis of metaphysical necessity/possibility is S5
 - Since the state of affairs are necessary beings, and at each world the same state of affairs exist (though different states of affairs obtain at different worlds), and since the same sets of

states of affairs form the maximally consistent sets at each world, it follows that the same set of possible worlds exist at each world. This entails that the accessibility relation is transitive, reflexive and symmetric

- This is useful to show that the selection of relational frame isn't dependent on concrete realism